Kandy Innovation Centre

National Institute of Business Management

UNDERGRADUATE RESEARCH PROPOSAL

Forecasting Sri Lankan Tourist Arrivals via Hybrid LSTM–XGBoost Ensemble Architectures Under Exogenous Shock Conditions

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1. Details of the Individual Research Project

1.1 Proposed Study Title

Forecasting Sri Lankan Tourist Arrivals via Hybrid LSTM–XGBoost Ensemble Architectures Under Exogenous Shock Conditions

1.2 Details of the Researcher

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2. Introduction

2.1 Background Information

Tourism is one of Sri Lanka's principal sources of foreign exchange revenue, directly supporting hotels, transport operators, airlines and allied small businesses. Government agencies and private operators must plan capacity, staffing and marketing spending months in advance, which requires an accurate forecast of monthly tourist arrivals. However, Sri Lanka's tourism sector has, within the last decade alone, experienced three severe exogenous shocks: the 2019 Easter Sunday terror attacks, the 2020–2021 COVID-19 pandemic, and the 2022 economic crisis and associated fuel shortages. Each of these events produced sharp, non-seasonal collapses in arrivals that classical forecasting approaches struggle to anticipate or recover quickly. Indeed, recent analyses of the COVID-19 pandemic demonstrate that forecasting models must be fundamentally modified to account for extreme data conditions, as traditional techniques become highly vulnerable during unprecedented global events (Rianda & Usman, 2023).

Classical time-series forecasting in tourism has historically relied on ARIMA and its seasonal variant, SARIMA, which model a series as a linear function of its own past values and past forecast errors (Box & Jenkins, 1970). These methods are well understood and computationally inexpensive, but they assume approximately linear, stationary relationships and are known to degrade sharply when a series experiences abrupt structural breaks. Structural breaks and location shifts are the primary culprits of systematic forecast failure in classical models, as they incorrectly assume that past relationships between variables will act as a reliable guide to the future (Castle, Clements & Hendry, 2016). Consequently, as noted by Hewapathirana (2023), SARIMA models trained on pre-2019 Sri Lankan tourist arrival data fail to accurately capture the profound impacts of the subsequent economic downturns and pandemic shocks, limiting their real-world utility in the current climate.

Recurrent neural architectures, particularly Long Short-Term Memory (LSTM) networks, were introduced to overcome the vanishing-gradient limitations of earlier recurrent models and are now widely used for sequence forecasting because they can retain long-range temporal dependencies such as yearly seasonality (Hochreiter & Schmidhuber, 1997). Separately, gradient-boosted tree ensembles such as XGBoost have become a leading method for structured/tabular prediction problems owing to their ability to model non-linear feature interactions efficiently (Chen & Guestrin, 2016). Hybrid architecture that combines a sequence model with a boosted-tree model — for instance, using the sequence model's output as an input

feature to the tree ensemble — have shown promise in other forecasting domains by capturing both temporal pattern and non-linear covariate effects simultaneously. Empirical evidence from financial time-series confirms that combining XGBoost to extract important features and an LSTM network to capture stochasticity significantly outperforms baseline ARIMA models (Vuong et al., 2022). Furthermore, hybrid approaches that merge linear and non-linear capabilities successfully cancel out individual model errors, leading to highly robust predictions even when tracking dynamic shifts (Ouassou & Taya, 2022).

Tourism demand modelling specifically has been surveyed extensively by Song, Witt and Li (2008), who note that forecast accuracy during crisis periods is consistently one of the weakest points of existing models and call for further work on crisis-robust forecasting methods. The Diebold–Mariano test (Diebold & Mariano, 1995) provides a recognized statistical procedure for comparing predictive accuracy between competing forecast models and will be adopted in this study to formally test whether the proposed hybrid ensemble improves on the classical baseline. To the researcher's knowledge, no existing published study has evaluated a stacked LSTM–XGBoost ensemble against SARIMA specifically on Sri Lankan tourist arrival data, nor explicitly segmented evaluation into stable versus shock regimes with formal statistical significance testing.

2.2 Research Problem

Sri Lanka Tourism Development Authority (SLTDA) forecasting practice currently relies primarily on historical averages and classical statistical models. While adequate under stable conditions, these approaches degrade sharply in accuracy during shock periods — precisely when accurate forecasts matter most for national economic planning. Traditional statistical forecasting methods often fail to capture the nonlinear, seasonal, and multifactor dynamics that characterize modern tourism data, making them particularly brittle during periods of structural breaks (Alsulami et al., 2026). There is therefore a clear, practically significant problem: no locally-validated forecasting method currently demonstrates robust accuracy for Sri Lankan tourist arrivals across both stable and crisis conditions, and no study has empirically tested whether a hybrid deep learning–ensemble learning architecture can close this gap.

2.3 Rationale

Given that Sri Lanka has experienced three major exogenous shocks within a single decade, and that further disruptive events (health, economic or geopolitical) cannot be ruled out, a forecasting method that remains reliable under both stable and shock conditions has direct value to SLTDA, the Ministry of Tourism, and private hospitality operators for capacity, staffing and marketing planning. As Castle, Clements and Hendry (2016) demonstrate, best in-sample fit is no guarantee of forecast success when shifts can occur; therefore, developing actively robust forecasting devices is essential. Beyond its local application, the project also contributes empirical evidence to the wider tourism-forecasting literature on whether hybrid ensemble architectures are more robust than single models under crisis conditions — directly addressing the gap identified by Song, Witt and Li (2008).

3. Objectives

3.1 Primary / Overall Objective

To design, implement, and evaluate a hybrid LSTM–XGBoost ensemble model for monthly Sri Lankan tourist arrivals that outperforms a classical SARIMA baseline in forecasting accuracy during exogenous shock periods.

3.2 Specific Research Objectives

1. To compile and clean a monthly time-series dataset of Sri Lankan tourist arrivals (2015–2025), integrating USD/LKR exchange-rate data, Google Trends search indices, and a documented shock-period flag.
2. To implement a SARIMA baseline alongside standalone LSTM and XGBoost models under identical preprocessing conditions to establish comparative performance benchmarks.
3. To design a stacked hybrid LSTM–XGBoost ensemble where the sequence model's predictions are passed as input features to the tree ensemble.
4. To evaluate and compare all models separately on stable-period and shock-period test data using RMSE, MAE, and MAPE metrics, supported by Diebold–Mariano significance testing, and to apply SHAP-based explainability to identify which features drive the ensemble's forecasts during shock periods.

4. Proposed Research Methods

4.1 General Description of the Study Field

This is a constructive (design-and-build) research project within the field of applied time-series forecasting and machine learning. A novel technical artefact — the hybrid LSTM–XGBoost ensemble — is designed, implemented and evaluated against classical and standalone-model baselines, consistent with common research designs for Computer Science / Artificial Intelligence final-year projects.

4.2 Research Questions

“Does a hybrid LSTM–XGBoost ensemble model forecast monthly Sri Lankan tourist arrivals with lower error (RMSE, MAE, MAPE) than a SARIMA baseline, both during stable periods and during exogenous shock periods (2019 Easter attacks; 2020–2022 COVID-19 pandemic; 2022 economic crisis), using SLTDA arrival data combined with USD/LKR exchange-rate and search-trend indicators?”

No formal statistical hypothesis of population parameters is tested; instead, the Diebold–Mariano test is used to assess whether the observed difference in forecast accuracy between the hybrid ensemble and the SARIMA baseline is statistically significant, separately for stable and shock-flagged periods.

4.3 Methods for Data Collection (linked to Objective 1)

The primary dataset is the SLTDA Monthly Tourist Arrivals Reports (2015–2025), a public government publication issued by the Sri Lanka Tourism Development Authority. This will be augmented with monthly-averaged USD/LKR exchange-rate data published by the Central Bank of Sri Lanka, and a monthly search-interest index for Sri Lanka-related travel queries obtained from Google Trends. The integration of web search data is highly supported by recent forecasting literature; Havranek and Zeynalov (2018) note that Google Trends information, particularly one week to two months ahead of arrivals, provides significant forecasting improvements and acts as a powerful concurrent indicator of tourist intent.

To ensure the search data accurately reflects inbound intent to Sri Lanka, specific keyword tracking will follow the methodology validated by Hewapathirana (2023), extracting monthly indices for "Sri Lanka Flights", "Sri Lanka Hotels", and "Sri Lanka Visa". A binary shock-period flag will be constructed manually from documented crisis timelines (April 2019 Easter attacks; March 2020–2021 COVID-19 pandemic; 2022 economic crisis and fuel-shortage

period). All three sources are publicly available and require no participant recruitment or primary data collection from human subjects.

4.4 Methods for Analyses of Data (linked to Objectives 2, 3 and 4)

A SARIMA model will serve as the classical statistical baseline. An LSTM network will be trained on lagged sequences of arrivals and exogenous features to capture seasonal and temporal dependencies. An XGBoost model will be trained on the same engineered features in tabular form. The hybrid ensemble will use a stacking approach, in which the LSTM's predicted value is supplied as an additional input feature to the XGBoost model. This architecture draws on the proven capability of XGBoost to handle high-dimensional, nonlinear residual error correction, while the LSTM excels at long-term temporal feature extraction (Vuong et al., 2022).

Data will be split chronologically (train: 2015–2021; test: 2022–2025) to avoid look-ahead bias, applying a moving window out-of-sample forecasting technique to rigorously validate the models against evolving data conditions (Hewapathirana, 2023). All models will be evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). As established in tourism demand literature, these metrics effectively capture both level accuracy and directional accuracy across different temporal phases (Sun et al., 2019; Alsulami et al., 2026). These metrics will be calculated separately for shock-flagged months and stable months. Additionally, SHAP values will be computed for the XGBoost component to interpret feature contributions during shock versus stable periods.

4.5 Methods for Implementation of the Solution (linked to Objective 3)

The ensemble will be implemented in Python, using standard open-source libraries (e.g. statsmodels/pmdarima for SARIMA, TensorFlow/Keras for the LSTM component, and the XGBoost library for the tree-ensemble component). Because deep learning algorithms rely crucially on the scale of inputs, all features will undergo Min-Max scaling to a range of 0 to 1 prior to LSTM training to optimize approximation and prevent vanishing gradients (Ouassou & Taya, 2022).

The stacked architecture will be built as a two-stage pipeline: the LSTM is trained first on the sequence data, its held-out predictions are then appended as a feature column, and the XGBoost model is trained on the combined tabular feature set. To avoid overfitting—a common issue when deploying complex networks on limited monthly observations—hyperparameter tuning will be executed utilizing GridSearchCV to identify the optimal configuration of decision trees, learning rates, and tree depths (Hewapathirana, 2023). The complete pipeline, trained models

and evaluation code will be maintained in a version-controlled code repository submitted alongside the final report.

4.6 Methods of Comparing Models and Results

Model comparison will proceed on two axes: (i) overall accuracy across the full test period, and (ii) accuracy conditioned on regime (stable vs. shock-flagged months), using RMSE, MAE and MAPE in both cases. The Diebold–Mariano test will be applied to the hybrid ensemble's forecast errors against the SARIMA baseline's forecast errors to determine whether the accuracy difference is statistically significant, computed separately for the stable and shock subsets. This regime-segmented comparison directly targets the project's central question of whether hybrid architectures are more robust than single models under crisis conditions.

Expected Results and Applications

- Objective 1 → A cleaned, reusable monthly Sri Lankan tourism time-series dataset with documented shock annotations, exchange-rate and search-trend features.
- Objective 2 → Working SARIMA, standalone LSTM and standalone XGBoost baseline models with recorded accuracy metrics.
- Objective 3 → A working hybrid LSTM–XGBoost ensemble forecasting system.
- Objective 4 → Statistically-tested empirical evidence (RMSE/MAE/MAPE + Diebold–Mariano) on whether the hybrid ensemble outperforms SARIMA overall and during shocks, together with SHAP-based explanations of which features drive forecast changes during crises.

These outcomes are of direct practical value to SLTDA, the Ministry of Tourism and private hospitality operators for capacity and marketing planning, and contribute a novel, locally grounded application of hybrid ensemble forecasting to the wider tourism-forecasting literature.

4.7 Resources

The project requires: (i) a personal computer with access to a Python data-science environment (or a cloud-based notebook service such as Google Colab) for model training; (ii) internet access to the SLTDA, Central Bank of Sri Lanka and Google Trends public data sources; and (iii) supervisor access for methodological guidance. No specialized laboratory space, proprietary software licenses, additional personnel or hardware purchases are required. All resources listed are already available to the researcher without further approval.

Ethics Approval

As the project uses only publicly published government and open-web statistical data (SLTDA, Central Bank of Sri Lanka, Google Trends) and involves no human participants, no personal, sensitive or personally identifiable information is collected. The completed university ethics self-assessment form will be submitted alongside this proposal (see Annexure), and no data collection beyond the public sources listed above will begin before ethics approval is confirmed. A dual-use risk is considered minimal, as the artefact is a demand-forecasting tool rather than one capable of being repurposed for harmful application.

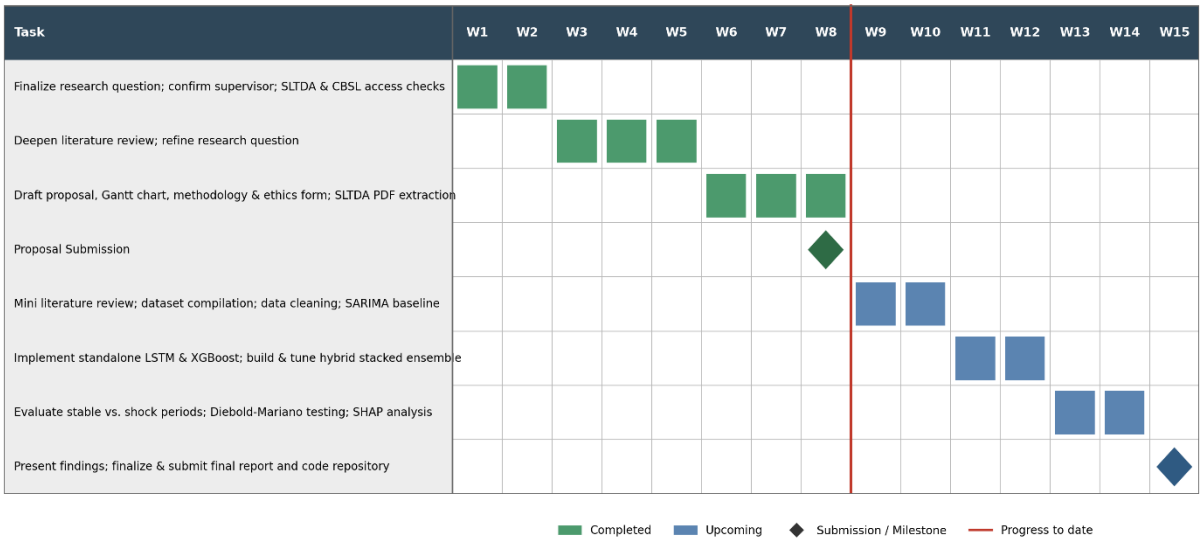
5. **Timeline**

The project is scheduled across both semesters of the Individual Project modules. Table 1 summarizes the planned phases in Gantt-chart form.

Table 1: Project Gantt / Phase Plan

Phase	**Planned Activity**
Weeks 1–2	Finalize research question; contact/confirm supervisor; begin SLTDA & CBSL data access checks.
Weeks 3–5	Deepen literature review (LSTM, XGBoost, hybrid ensembles, tourism demand forecasting); refine research question.
Weeks 6–8	Draft proposal, Gantt chart, methodology and ethics form; begin manual/automated extraction of SLTDA monthly arrival PDFs. Completed Proposal Submission Submit project proposal, Gantt chart, selected methodology and ethics form.
Weeks 9–12	Conduct mini literature review (12–15 sources); finalize raw dataset compilation (arrivals, exchange rate, shock labels); data cleaning and feature engineering; implement SARIMA baseline.
Weeks 11–12	**Implement standalone LSTM and XGBoost models; build and tune the hybrid stacked ensemble**
Weeks 13–14	Run controlled evaluation across stable vs. shock periods; Diebold–Mariano significance testing; SHAP analysis.
Week 15	Present findings and working system; finalize, revise and submit final report and code repository.

15-Week Project Gantt Chart: Hybrid LSTM-XGBoost Tourism Forecasting Project



6. Status of Permits and Clearance

No permits from external regulatory bodies are required for this project. Because the study relies entirely on publicly available secondary statistical data — SLTDA arrivals reports, Central Bank of Sri Lanka exchange rates, and Google Trends search indices — and involves no human participants, animal subjects, or sensitive private datasets, no special data-gathering permits apply. The only requisite approval is institutional ethical clearance from the university, which will be secured through the standard internal application process; the completed ethics self-assessment form will be submitted alongside this proposal (see Annexure), and no data processing beyond the public sources listed above will begin before that approval is confirmed.

7. References

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8. Annexure

Ethical Clearance Application

The completed NIBM Ethical Approval Application Form for this project is attached as a separate document alongside this proposal. As this study relies solely on publicly available secondary data (SLTDA monthly tourist arrivals reports, Central Bank of Sri Lanka exchange-rate data, and Google Trends search-interest indices) and involves no human participants, the form has been completed on that basis, with the participant-related sections marked not applicable accordingly. The form has been signed by the researcher and supervisor as required.